

Project Title

Match Outcome Prediction Using Logistic Regression and Random Forest

Project Report submitted for the partial fulfilment of grade for the course

Machine learning (25SC2107E)

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ABSTRACT

This project presents a Match Outcome Prediction System using Logistic Regression and Random Forest algorithms to forecast the result of sports matches based on historical performance data. Traditional manual analysis of team statistics is time-consuming and often fails to capture complex patterns hidden in large datasets. The proposed system overcomes this limitation by training two machine learning models—Logistic Regression and Random Forest—on features such as team form, home/away advantage, past head-to-head records, and average scoring performance. These models classify the outcome of a match as a win or loss and their predictions are compared to identify the more accurate approach. The system evaluates model performance using accuracy, precision, recall, and confusion matrices, providing a clear comparison between a simple linear model and an ensemble-based model. The application demonstrates the practical use of machine learning in sports analytics, enabling data-driven insights for outcome forecasting, fan engagement platforms, and sports broadcasting analysis. The project highlights how predictive modeling can enhance decision-making in the sports and entertainment industry.

Signature of the Guide